

Variability Analysis: MESLM, T-Test, Pitman-Morgan (H1b, H2b, H3b)

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```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.1.4      v readr      2.1.5
## v forcats    1.0.1      v stringr   1.5.2
## v ggplot2    4.0.0      v tibble    3.3.0
## v lubridate  1.9.4      v tidyr     1.3.1
## v purrr      1.1.0
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(PairedData)
```

```
## Loading required package: MASS
##
## Attaching package: 'MASS'
##
## The following object is masked from 'package:dplyr':
##
##     select
##
## Loading required package: gld
## Loading required package: mvtnorm
## Loading required package: lattice
##
## Attaching package: 'PairedData'
##
## The following object is masked from 'package:base':
##
##     summary
```

```
library(brms)
```

```
## Loading required package: Rcpp
## Loading 'brms' package (version 2.23.0). Useful instructions
```

```
## can be found by typing help('brms'). A more detailed introduction
## to the package is available through vignette('brms_overview').
##
## Attaching package: 'brms'
##
## The following object is masked from 'package:stats':
##
##     ar
```

```
# Load Interleaf data
df <- read.csv("../Data/df_interleaf_measures.csv")
```

```
# T-Test: Within participant
# Aggregate Data: Calculate SD per condition per participant
df_sd <- df %>%
  group_by(participant_ID, shared_condition_order) %>%
  summarise(
    sd_flow = sd(flow_rating),
    sd_social_presence = sd(social_presence_rating),
    sd_affect = sd(panas_rating),
    .groups = "drop"
  ) %>%
  pivot_wider(names_from = shared_condition_order,
              values_from = c(sd_flow, sd_social_presence, sd_affect))
```

```
head(df_sd)
```

```
## # A tibble: 6 x 7
##   participant_ID      sd_flow_with_ai sd_flow_without_ai sd_social_presence_w-1
##   <chr>              <dbl>              <dbl>              <dbl>
## 1 participant_chattin~      31.6              9.57              19.7
## 2 participant_chattin~      12.3              0.816            34.5
## 3 participant_chattin~       1.89              2.87              8.18
## 4 participant_chattin~      10.4              15.1             12.2
## 5 participant_chattin~       8.42              22.3              2.5
## 6 participant_chattin~      10.0              17.0             14.4
## # i abbreviated name: 1: sd_social_presence_with_ai
## # i 3 more variables: sd_social_presence_without_ai <dbl>,
## #   sd_affect_with_ai <dbl>, sd_affect_without_ai <dbl>
```

```
# T-Tests: Comparing with and without sd within participant
t.test(df_sd$sd_flow_without_ai, df_sd$sd_flow_with_ai, paired = TRUE) # Flow
```

```
##
## Paired t-test
##
## data: df_sd$sd_flow_without_ai and df_sd$sd_flow_with_ai
## t = 2.3272, df = 41, p-value = 0.02497
## alternative hypothesis: true mean difference is not equal to 0
## 95 percent confidence interval:
##  0.5860888 8.2802960
## sample estimates:
```

```
## mean difference
##      4.433192
```

```
t.test(df_sd$sd_social_presence_without_ai, df_sd$sd_social_presence_with_ai, paired = TRUE) # Social P
```

```
##
## Paired t-test
##
## data: df_sd$sd_social_presence_without_ai and df_sd$sd_social_presence_with_ai
## t = 2.2661, df = 41, p-value = 0.02879
## alternative hypothesis: true mean difference is not equal to 0
## 95 percent confidence interval:
##    0.5883825 10.2278854
## sample estimates:
## mean difference
##      5.408134
```

```
t.test(df_sd$sd_affect_without_ai, df_sd$sd_affect_with_ai, paired = TRUE) # Affect
```

```
##
## Paired t-test
##
## data: df_sd$sd_affect_without_ai and df_sd$sd_affect_with_ai
## t = 1.4608, df = 41, p-value = 0.1517
## alternative hypothesis: true mean difference is not equal to 0
## 95 percent confidence interval:
##   -0.6868844  4.2785452
## sample estimates:
## mean difference
##      1.79583
```

```
# Pitman Morgan: Variance test Between participants
# Aggregate Data: Calculate mean per condition per participant
```

```
df_mean <- df %>%
  group_by(participant_ID, shared_condition_order) %>%
  summarise(
    mean_flow = mean(flow_rating),
    mean_social_presence = mean(social_presence_rating),
    mean_affect = mean(panas_rating),
    .groups = "drop"
  ) %>%
  pivot_wider(names_from = shared_condition_order,
              values_from = c(mean_flow, mean_social_presence, mean_affect))
```

```
# Pitman-Morgan: Comparing with-ai and without-ai mean, sd between participant
Var.test(df_mean$mean_flow_without_ai, df_mean$mean_flow_with_ai, paired = TRUE) # Flow
```

```
##
## Paired Pitman-Morgan test
##
## data: x and y
```

```

## t = -2.1621, df = 40, p-value = 0.03665
## alternative hypothesis: true ratio of variances is not equal to 1
## 95 percent confidence interval:
## 0.3059884 0.9619437
## sample estimates:
## variance of x variance of y
##      326.5916      601.9740

Var.test(df_mean$mean_social_presence_without_ai, df_mean$mean_social_presence_with_ai, paired = TRUE)

##
## Paired Pitman-Morgan test
##
## data: x and y
## t = -3.4284, df = 40, p-value = 0.00142
## alternative hypothesis: true ratio of variances is not equal to 1
## 95 percent confidence interval:
## 0.1961857 0.6702303
## sample estimates:
## variance of x variance of y
##      265.2019      731.3596

Var.test(df_mean$mean_affect_without_ai, df_mean$mean_affect_with_ai, paired = TRUE) # Affect

##
## Paired Pitman-Morgan test
##
## data: x and y
## t = -2.8518, df = 40, p-value = 0.006847
## alternative hypothesis: true ratio of variances is not equal to 1
## 95 percent confidence interval:
## 0.3850477 0.8523426
## sample estimates:
## variance of x variance of y
##      206.2372      360.0002

# Sum contrasts: with_ai = -0.5, without_ai = 0.5
df$shared_condition_order <- factor(df$shared_condition_order,
                                   levels = c("with_ai", "without_ai"))
contrasts(df$shared_condition_order) <- c(-0.5, 0.5)

# Print Contrast Matrix
contrasts(df$shared_condition_order)

##           [,1]
## with_ai    -0.5
## without_ai  0.5

# Check that everything is there and working
attributes(df$shared_condition_order)

```

```
## $levels
## [1] "with_ai"      "without_ai"
##
## $class
## [1] "factor"
##
## $contrasts
##           [,1]
## with_ai    -0.5
## without_ai  0.5
```

```
df <- df %>% # Create a col for each condition, to allow different intercepts per condition
  mutate(
    condition_hai = as.integer(shared_condition_order == "with_ai"),
    condition_hh = as.integer(shared_condition_order == "without_ai")
  )
```

```
df
```

```
##           participant_ID shared_prompt_order
## 1 participant_chatting_castellum_693305a50416fb7a91dcf68a 1
## 2 participant_chatting_castellum_693305a50416fb7a91dcf68a 3
## 3 participant_chatting_castellum_693305a50416fb7a91dcf68a 2
## 4 participant_chatting_castellum_693305a50416fb7a91dcf68a 4
## 5 participant_chatting_castellum_693305a50416fb7a91dcf68a 1
## 6 participant_chatting_castellum_693305a50416fb7a91dcf68a 4
## 7 participant_chatting_castellum_693305a50416fb7a91dcf68a 2
## 8 participant_chatting_castellum_693305a50416fb7a91dcf68a 3
## 9 participant_chatting_castellum_693306e60416fb7a91dcf68e 1
## 10 participant_chatting_castellum_693306e60416fb7a91dcf68e 3
## 11 participant_chatting_castellum_693306e60416fb7a91dcf68e 2
## 12 participant_chatting_castellum_693306e60416fb7a91dcf68e 4
## 13 participant_chatting_castellum_693306e60416fb7a91dcf68e 1
## 14 participant_chatting_castellum_693306e60416fb7a91dcf68e 4
## 15 participant_chatting_castellum_693306e60416fb7a91dcf68e 2
## 16 participant_chatting_castellum_693306e60416fb7a91dcf68e 3
## 17 participant_chatting_castellum_695cf5570416fb7a91dd18bd 1
## 18 participant_chatting_castellum_695cf5570416fb7a91dd18bd 3
## 19 participant_chatting_castellum_695cf5570416fb7a91dd18bd 3
## 20 participant_chatting_castellum_695cf5570416fb7a91dd18bd 2
## 21 participant_chatting_castellum_695cf5570416fb7a91dd18bd 4
## 22 participant_chatting_castellum_695cf5570416fb7a91dd18bd 4
## 23 participant_chatting_castellum_695cf5570416fb7a91dd18bd 1
## 24 participant_chatting_castellum_695cf5570416fb7a91dd18bd 2
## 25 participant_chatting_castellum_695cf56a0416fb7a91dd18bf 1
## 26 participant_chatting_castellum_695cf56a0416fb7a91dd18bf 3
## 27 participant_chatting_castellum_695cf56a0416fb7a91dd18bf 3
## 28 participant_chatting_castellum_695cf56a0416fb7a91dd18bf 2
## 29 participant_chatting_castellum_695cf56a0416fb7a91dd18bf 4
## 30 participant_chatting_castellum_695cf56a0416fb7a91dd18bf 4
## 31 participant_chatting_castellum_695cf56a0416fb7a91dd18bf 1
## 32 participant_chatting_castellum_695cf56a0416fb7a91dd18bf 2
## 33 participant_chatting_castellum_695d1b780416fb7a91dd195a 1
## 34 participant_chatting_castellum_695d1b780416fb7a91dd195a 1
```

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

## 305	participant_chatting_castellum_698718b00416fb7a91dd2716	1
## 306	participant_chatting_castellum_698718b00416fb7a91dd2716	3
## 307	participant_chatting_castellum_698718b00416fb7a91dd2716	2
## 308	participant_chatting_castellum_698718b00416fb7a91dd2716	4
## 309	participant_chatting_castellum_698718b00416fb7a91dd2716	1
## 310	participant_chatting_castellum_698718b00416fb7a91dd2716	4
## 311	participant_chatting_castellum_698718b00416fb7a91dd2716	2
## 312	participant_chatting_castellum_698718b00416fb7a91dd2716	3
## 313	participant_chatting_castellum_6987210b0416fb7a91dd2718	1
## 314	participant_chatting_castellum_6987210b0416fb7a91dd2718	3
## 315	participant_chatting_castellum_6987210b0416fb7a91dd2718	2
## 316	participant_chatting_castellum_6987210b0416fb7a91dd2718	4
## 317	participant_chatting_castellum_6987210b0416fb7a91dd2718	1
## 318	participant_chatting_castellum_6987210b0416fb7a91dd2718	4
## 319	participant_chatting_castellum_6987210b0416fb7a91dd2718	2
## 320	participant_chatting_castellum_6987210b0416fb7a91dd2718	3
## 321	participant_chatting_castellum_698737b50416fb7a91dd27c0	2
## 322	participant_chatting_castellum_698737b50416fb7a91dd27c0	1
## 323	participant_chatting_castellum_698737b50416fb7a91dd27c0	1
## 324	participant_chatting_castellum_698737b50416fb7a91dd27c0	3
## 325	participant_chatting_castellum_698737b50416fb7a91dd27c0	3
## 326	participant_chatting_castellum_698737b50416fb7a91dd27c0	2
## 327	participant_chatting_castellum_698737b50416fb7a91dd27c0	4
## 328	participant_chatting_castellum_698737b50416fb7a91dd27c0	4
## 329	participant_chatting_castellum_698737d00416fb7a91dd27c2	2
## 330	participant_chatting_castellum_698737d00416fb7a91dd27c2	1
## 331	participant_chatting_castellum_698737d00416fb7a91dd27c2	1
## 332	participant_chatting_castellum_698737d00416fb7a91dd27c2	3
## 333	participant_chatting_castellum_698737d00416fb7a91dd27c2	3
## 334	participant_chatting_castellum_698737d00416fb7a91dd27c2	2
## 335	participant_chatting_castellum_698737d00416fb7a91dd27c2	4
## 336	participant_chatting_castellum_698737d00416fb7a91dd27c2	4
##	chat_number shared_condition_order dyad_ID	
## 1	1 with_ai dyad_4532a640	
## 2	2 with_ai dyad_4532a640	
## 3	3 without_ai dyad_4532a640	
## 4	4 with_ai dyad_4532a640	
## 5	5 without_ai dyad_4532a640	
## 6	6 without_ai dyad_4532a640	
## 7	7 with_ai dyad_4532a640	
## 8	8 without_ai dyad_4532a640	
## 9	1 with_ai dyad_4532a640	
## 10	2 with_ai dyad_4532a640	
## 11	3 without_ai dyad_4532a640	
## 12	4 with_ai dyad_4532a640	
## 13	5 without_ai dyad_4532a640	
## 14	6 without_ai dyad_4532a640	
## 15	7 with_ai dyad_4532a640	
## 16	8 without_ai dyad_4532a640	
## 17	1 without_ai dyad_ff8fbb6f	
## 18	2 with_ai dyad_ff8fbb6f	
## 19	3 without_ai dyad_ff8fbb6f	
## 20	4 without_ai dyad_ff8fbb6f	
## 21	5 without_ai dyad_ff8fbb6f	

## 22	6	with_ai dyad_ff8fbb6f
## 23	7	with_ai dyad_ff8fbb6f
## 24	8	with_ai dyad_ff8fbb6f
## 25	1	without_ai dyad_ff8fbb6f
## 26	2	with_ai dyad_ff8fbb6f
## 27	3	without_ai dyad_ff8fbb6f
## 28	4	without_ai dyad_ff8fbb6f
## 29	5	without_ai dyad_ff8fbb6f
## 30	6	with_ai dyad_ff8fbb6f
## 31	7	with_ai dyad_ff8fbb6f
## 32	8	with_ai dyad_ff8fbb6f
## 33	1	without_ai dyad_61225b7b
## 34	2	with_ai dyad_61225b7b
## 35	3	without_ai dyad_61225b7b
## 36	4	without_ai dyad_61225b7b
## 37	5	with_ai dyad_61225b7b
## 38	6	with_ai dyad_61225b7b
## 39	7	with_ai dyad_61225b7b
## 40	8	without_ai dyad_61225b7b
## 41	1	without_ai dyad_61225b7b
## 42	2	with_ai dyad_61225b7b
## 43	3	without_ai dyad_61225b7b
## 44	4	without_ai dyad_61225b7b
## 45	5	with_ai dyad_61225b7b
## 46	6	with_ai dyad_61225b7b
## 47	7	with_ai dyad_61225b7b
## 48	8	without_ai dyad_61225b7b
## 49	1	with_ai dyad_8d1bdca5
## 50	2	without_ai dyad_8d1bdca5
## 51	3	with_ai dyad_8d1bdca5
## 52	4	with_ai dyad_8d1bdca5
## 53	5	without_ai dyad_8d1bdca5
## 54	6	with_ai dyad_8d1bdca5
## 55	7	without_ai dyad_8d1bdca5
## 56	8	without_ai dyad_8d1bdca5
## 57	1	with_ai dyad_8d1bdca5
## 58	2	without_ai dyad_8d1bdca5
## 59	3	with_ai dyad_8d1bdca5
## 60	4	with_ai dyad_8d1bdca5
## 61	5	without_ai dyad_8d1bdca5
## 62	6	with_ai dyad_8d1bdca5
## 63	7	without_ai dyad_8d1bdca5
## 64	8	without_ai dyad_8d1bdca5
## 65	1	with_ai dyad_c4e7f1c9
## 66	2	with_ai dyad_c4e7f1c9
## 67	3	with_ai dyad_c4e7f1c9
## 68	4	without_ai dyad_c4e7f1c9
## 69	5	without_ai dyad_c4e7f1c9
## 70	6	without_ai dyad_c4e7f1c9
## 71	7	with_ai dyad_c4e7f1c9
## 72	8	without_ai dyad_c4e7f1c9
## 73	1	with_ai dyad_c4e7f1c9
## 74	2	with_ai dyad_c4e7f1c9
## 75	3	with_ai dyad_c4e7f1c9

## 76	4	without_ai dyad_c4e7f1c9
## 77	5	without_ai dyad_c4e7f1c9
## 78	6	without_ai dyad_c4e7f1c9
## 79	7	with_ai dyad_c4e7f1c9
## 80	8	without_ai dyad_c4e7f1c9
## 81	1	without_ai dyad_82824864
## 82	2	with_ai dyad_82824864
## 83	3	without_ai dyad_82824864
## 84	4	with_ai dyad_82824864
## 85	5	with_ai dyad_82824864
## 86	6	without_ai dyad_82824864
## 87	7	without_ai dyad_82824864
## 88	8	with_ai dyad_82824864
## 89	1	without_ai dyad_82824864
## 90	2	with_ai dyad_82824864
## 91	3	without_ai dyad_82824864
## 92	4	with_ai dyad_82824864
## 93	5	with_ai dyad_82824864
## 94	6	without_ai dyad_82824864
## 95	7	without_ai dyad_82824864
## 96	8	with_ai dyad_82824864
## 97	1	with_ai dyad_a9880858
## 98	2	without_ai dyad_a9880858
## 99	3	with_ai dyad_a9880858
## 100	4	with_ai dyad_a9880858
## 101	5	without_ai dyad_a9880858
## 102	6	without_ai dyad_a9880858
## 103	7	without_ai dyad_a9880858
## 104	8	with_ai dyad_a9880858
## 105	1	with_ai dyad_a9880858
## 106	2	without_ai dyad_a9880858
## 107	3	with_ai dyad_a9880858
## 108	4	with_ai dyad_a9880858
## 109	5	without_ai dyad_a9880858
## 110	6	without_ai dyad_a9880858
## 111	7	without_ai dyad_a9880858
## 112	8	with_ai dyad_a9880858
## 113	1	without_ai dyad_c4ef9844
## 114	2	without_ai dyad_c4ef9844
## 115	3	with_ai dyad_c4ef9844
## 116	4	with_ai dyad_c4ef9844
## 117	5	without_ai dyad_c4ef9844
## 118	6	with_ai dyad_c4ef9844
## 119	7	with_ai dyad_c4ef9844
## 120	8	without_ai dyad_c4ef9844
## 121	1	without_ai dyad_c4ef9844
## 122	2	without_ai dyad_c4ef9844
## 123	3	with_ai dyad_c4ef9844
## 124	4	with_ai dyad_c4ef9844
## 125	5	without_ai dyad_c4ef9844
## 126	6	with_ai dyad_c4ef9844
## 127	7	with_ai dyad_c4ef9844
## 128	8	without_ai dyad_c4ef9844
## 129	1	with_ai dyad_64f09910

## 130	2	with_ai dyad_64f09910
## 131	3	without_ai dyad_64f09910
## 132	4	with_ai dyad_64f09910
## 133	5	without_ai dyad_64f09910
## 134	6	without_ai dyad_64f09910
## 135	7	with_ai dyad_64f09910
## 136	8	without_ai dyad_64f09910
## 137	1	with_ai dyad_64f09910
## 138	2	with_ai dyad_64f09910
## 139	3	without_ai dyad_64f09910
## 140	4	with_ai dyad_64f09910
## 141	5	without_ai dyad_64f09910
## 142	6	without_ai dyad_64f09910
## 143	7	with_ai dyad_64f09910
## 144	8	without_ai dyad_64f09910
## 145	1	with_ai dyad_4be886de
## 146	2	with_ai dyad_4be886de
## 147	3	without_ai dyad_4be886de
## 148	4	without_ai dyad_4be886de
## 149	5	with_ai dyad_4be886de
## 150	6	without_ai dyad_4be886de
## 151	7	with_ai dyad_4be886de
## 152	8	without_ai dyad_4be886de
## 153	1	with_ai dyad_4be886de
## 154	2	with_ai dyad_4be886de
## 155	3	without_ai dyad_4be886de
## 156	4	without_ai dyad_4be886de
## 157	5	with_ai dyad_4be886de
## 158	6	without_ai dyad_4be886de
## 159	7	with_ai dyad_4be886de
## 160	8	without_ai dyad_4be886de
## 161	1	without_ai dyad_9ee16729
## 162	2	without_ai dyad_9ee16729
## 163	3	without_ai dyad_9ee16729
## 164	4	with_ai dyad_9ee16729
## 165	5	with_ai dyad_9ee16729
## 166	6	with_ai dyad_9ee16729
## 167	7	with_ai dyad_9ee16729
## 168	8	without_ai dyad_9ee16729
## 169	1	without_ai dyad_9ee16729
## 170	2	without_ai dyad_9ee16729
## 171	3	without_ai dyad_9ee16729
## 172	4	with_ai dyad_9ee16729
## 173	5	with_ai dyad_9ee16729
## 174	6	with_ai dyad_9ee16729
## 175	7	with_ai dyad_9ee16729
## 176	8	without_ai dyad_9ee16729
## 177	1	with_ai dyad_b1705d6c
## 178	2	without_ai dyad_b1705d6c
## 179	3	without_ai dyad_b1705d6c
## 180	4	with_ai dyad_b1705d6c
## 181	5	without_ai dyad_b1705d6c
## 182	6	with_ai dyad_b1705d6c
## 183	7	without_ai dyad_b1705d6c

## 184	8	with_ai dyad_b1705d6c
## 185	1	with_ai dyad_b1705d6c
## 186	2	without_ai dyad_b1705d6c
## 187	3	without_ai dyad_b1705d6c
## 188	4	with_ai dyad_b1705d6c
## 189	5	without_ai dyad_b1705d6c
## 190	6	with_ai dyad_b1705d6c
## 191	7	without_ai dyad_b1705d6c
## 192	8	with_ai dyad_b1705d6c
## 193	1	without_ai dyad_5609739b
## 194	2	without_ai dyad_5609739b
## 195	3	with_ai dyad_5609739b
## 196	4	with_ai dyad_5609739b
## 197	5	without_ai dyad_5609739b
## 198	6	with_ai dyad_5609739b
## 199	7	with_ai dyad_5609739b
## 200	8	without_ai dyad_5609739b
## 201	1	without_ai dyad_5609739b
## 202	2	without_ai dyad_5609739b
## 203	3	with_ai dyad_5609739b
## 204	4	with_ai dyad_5609739b
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## 336	participant_chatting_castellum_698737b50416fb7a91dd27c0	41
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Mir ist aufgefallen, dass ich den Gesprächsverlauf

Der Chat fühlte sich mehr nach einem Flow an. Mein Thema war zwar persönlich, aber nicht zu sehr

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## 336
##      ai_detection_rating social_presence_rating likability_rating time_submitted
## 1              88              28              19              6
## 2              78              63              68              1
## 3              9              68              59              2
## 4             87              74              78              2
## 5              8              29              50              6
## 6              5              28              48              1
## 7             88              59              50              8
## 8              0              65              50              6
## 9            100              0              49              3
## 10             19              69              50              4
## 11             0             100             100              3
## 12            100              0              50              4
## 13             0             100             100              2
## 14             0             100             100              4
## 15            100              0              50              2
## 16             0             100             100              2
## 17             0              39              14              2
## 18            100              9              4              3
## 19             0              39              19              6
## 20             0              39              9              3
## 21             0              29              9              3
## 22            100              29              18              2
## 23            100              19              14              3
## 24            100              18              19              4
## 25             18              58              50              6
## 26             9              78              58             15
## 27             8              39              63             11
## 28             79              13              33              4
## 29             77              39              38              2
## 30             81              66              62              7
## 31             89              85              85              7
## 32             99              95              79              3
## 33             22              29              60              7
## 34            100             100              41              2
## 35             30              78              74              2
## 36             39              61              78              3

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## 37	100	95	43	2
## 38	100	100	56	2
## 39	100	100	27	1
## 40	8	72	72	3
## 41	38	28	29	9
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## 43	29	50	59	9
## 44	19	50	39	4
## 45	100	49	19	3
## 46	100	39	19	3
## 47	100	38	28	15
## 48	19	29	39	3
## 49	100	9	49	3
## 50	29	49	49	2
## 51	100	8	9	3
## 52	0	89	83	7
## 53	13	69	59	4
## 54	100	9	9	3
## 55	0	85	62	1
## 56	0	85	73	4
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## 58	0	100	69	1
## 59	100	68	69	1
## 60	100	89	88	1
## 61	0	99	53	1
## 62	100	100	96	1
## 63	0	78	73	1
## 64	0	94	89	1
## 65	88	89	97	17
## 66	90	99	98	2
## 67	50	83	97	6
## 68	27	74	60	4
## 69	16	100	95	3
## 70	81	32	27	3
## 71	90	100	100	4
## 72	28	100	93	8
## 73	100	68	50	3
## 74	79	58	50	5
## 75	79	69	38	3
## 76	19	68	29	2
## 77	0	78	49	3
## 78	0	68	38	3
## 79	100	78	28	1
## 80	0	29	30	3
## 81	11	7	7	3
## 82	100	80	0	2
## 83	9	36	21	1
## 84	100	66	22	3
## 85	100	56	5	1
## 86	5	89	75	2
## 87	20	66	47	2
## 88	100	30	0	2
## 89	63	43	54	5
## 90	100	68	73	1

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## 93	100	64	54	2
## 94	0	64	69	2
## 95	0	64	63	1
## 96	99	58	44	3
## 97	90	60	19	19
## 98	50	30	25	4
## 99	88	68	29	4
## 100	39	40	50	3
## 101	20	69	69	3
## 102	16	66	72	2
## 103	17	75	68	5
## 104	94	28	11	5
## 105	89	23	39	11
## 106	19	39	48	1
## 107	94	29	14	2
## 108	100	18	19	4
## 109	9	59	59	2
## 110	9	29	48	1
## 111	9	47	59	2
## 112	100	38	38	2
## 113	100	19	23	1
## 114	33	79	47	1
## 115	100	70	35	2
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## 119	100	83	50	2
## 120	1	97	73	4
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## 126	100	58	7	2
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## 134	0	63	35	2
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## 136	0	73	61	3
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## 138	95	65	65	2
## 139	1	60	59	1
## 140	94	74	64	5
## 141	4	79	76	2
## 142	4	68	67	2
## 143	97	88	88	3
## 144	5	67	71	1

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## 150	24	64	66	7
## 151	96	6	10	7
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## 153	93	89	68	20
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## 156	13	60	50	4
## 157	94	70	71	4
## 158	13	68	66	5
## 159	91	78	69	4
## 160	8	79	70	2
## 161	21	88	93	8
## 162	62	68	92	3
## 163	0	93	96	2
## 164	100	100	63	2
## 165	100	31	7	2
## 166	100	83	11	2
## 167	100	87	61	1
## 168	0	100	100	1
## 169	8	79	89	1
## 170	88	89	89	1
## 171	84	58	82	1
## 172	100	100	77	2
## 173	100	100	76	3
## 174	100	100	88	2
## 175	100	100	74	1
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## 177	100	0	29	9
## 178	0	83	47	7
## 179	0	94	68	6
## 180	100	69	59	3
## 181	0	100	69	3
## 182	100	69	79	3
## 183	0	84	63	2
## 184	100	77	48	2
## 185	100	79	59	9
## 186	8	68	64	2
## 187	0	58	58	3
## 188	100	79	52	1
## 189	0	52	53	2
## 190	100	90	68	1
## 191	0	59	58	3
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## 193	79	18	18	3
## 194	68	50	18	3
## 195	100	79	38	3
## 196	100	79	68	13
## 197	0	50	40	3
## 198	100	89	79	5

## 199	100	79	50	2
## 200	0	29	48	5
## 201	8	84	69	3
## 202	38	64	60	1
## 203	88	38	48	1
## 204	87	38	48	2
## 205	28	73	63	13
## 206	88	50	50	3
## 207	87	39	48	2
## 208	27	64	63	2
## 209	8	18	48	2
## 210	100	100	48	2
## 211	99	100	48	3
## 212	100	100	48	2
## 213	0	100	88	2
## 214	0	61	70	2
## 215	100	89	50	2
## 216	0	100	97	1
## 217	79	50	77	1
## 218	100	88	94	4
## 219	100	95	94	2
## 220	100	88	94	3
## 221	99	69	79	3
## 222	89	60	69	1
## 223	100	96	59	21
## 224	84	74	89	48
## 225	8	69	89	10
## 226	100	100	100	11
## 227	100	95	100	3
## 228	18	88	100	1
## 229	19	100	100	7
## 230	100	100	100	1
## 231	0	79	100	5
## 232	100	100	100	3
## 233	10	71	64	25
## 234	100	100	88	8
## 235	100	89	99	44
## 236	0	68	64	10
## 237	0	77	84	41
## 238	100	84	100	10
## 239	0	74	74	15
## 240	100	97	96	17
## 241	50	11	48	1
## 242	100	82	69	2
## 243	89	69	74	2
## 244	9	30	21	2
## 245	13	75	75	2
## 246	99	81	64	2
## 247	11	79	65	2
## 248	98	78	16	4
## 249	0	95	77	12
## 250	100	91	89	5
## 251	100	100	99	4
## 252	0	100	92	3

## 253	0	96	89	1
## 254	100	100	100	3
## 255	0	95	94	3
## 256	100	100	99	2
## 257	0	19	9	7
## 258	8	68	39	3
## 259	100	100	18	2
## 260	100	89	38	2
## 261	0	79	48	2
## 262	100	89	50	2
## 263	100	78	58	2
## 264	0	69	69	4
## 265	39	69	49	2
## 266	31	79	50	10
## 267	100	0	9	2
## 268	100	69	49	5
## 269	5	63	59	20
## 270	100	80	66	1
## 271	100	69	49	2
## 272	0	22	54	5
## 273	99	4	9	5
## 274	100	99	30	2
## 275	89	87	50	2
## 276	100	79	56	2
## 277	100	89	56	2
## 278	0	100	91	2
## 279	0	100	88	2
## 280	0	89	79	2
## 281	16	28	28	2
## 282	93	34	44	4
## 283	87	59	35	2
## 284	94	59	55	2
## 285	91	65	53	2
## 286	9	86	69	2
## 287	17	87	85	3
## 288	1	94	99	2
## 289	89	28	17	2
## 290	100	39	28	2
## 291	100	58	33	2
## 292	4	88	77	1
## 293	28	33	27	6
## 294	89	34	28	1
## 295	100	48	19	1
## 296	100	50	26	2
## 297	53	50	58	2
## 298	94	95	63	2
## 299	95	95	69	2
## 300	23	75	90	2
## 301	15	83	93	1
## 302	20	70	89	1
## 303	91	87	85	2
## 304	95	87	84	1
## 305	100	100	100	3
## 306	0	58	12	2

## 307	0	64	50	2
## 308	0	29	22	2
## 309	0	58	56	5
## 310	100	100	100	2
## 311	100	100	100	2
## 312	100	100	100	3
## 313	100	57	50	2
## 314	0	68	47	2
## 315	0	60	49	6
## 316	0	78	59	6
## 317	9	68	64	3
## 318	100	81	45	3
## 319	0	88	41	4
## 320	0	80	41	3
## 321	19	88	79	2
## 322	68	28	49	2
## 323	100	9	0	2
## 324	48	58	59	2
## 325	100	0	0	2
## 326	100	39	0	3
## 327	79	58	59	1
## 328	100	9	0	2
## 329	54	79	78	5
## 330	33	59	79	1
## 331	100	32	49	2
## 332	0	84	75	1
## 333	100	38	50	1
## 334	100	6	19	2
## 335	0	93	88	2
## 336	100	16	37	1
##	panas_rating	condition_hai	condition_hh	
## 1	50	1	0	
## 2	69	1	0	
## 3	50	0	1	
## 4	78	1	0	
## 5	50	0	1	
## 6	40	0	1	
## 7	68	1	0	
## 8	50	0	1	
## 9	50	1	0	
## 10	100	1	0	
## 11	49	0	1	
## 12	50	1	0	
## 13	50	0	1	
## 14	79	0	1	
## 15	80	1	0	
## 16	100	0	1	
## 17	48	0	1	
## 18	39	1	0	
## 19	29	0	1	
## 20	28	0	1	
## 21	28	0	1	
## 22	34	1	0	
## 23	28	1	0	

## 24	29	1	0
## 25	48	0	1
## 26	50	1	0
## 27	50	0	1
## 28	41	0	1
## 29	48	0	1
## 30	62	1	0
## 31	84	1	0
## 32	71	1	0
## 33	79	0	1
## 34	80	1	0
## 35	81	0	1
## 36	79	0	1
## 37	79	1	0
## 38	81	1	0
## 39	68	1	0
## 40	71	0	1
## 41	39	0	1
## 42	59	1	0
## 43	50	0	1
## 44	49	0	1
## 45	57	1	0
## 46	38	1	0
## 47	38	1	0
## 48	29	0	1
## 49	48	1	0
## 50	48	0	1
## 51	50	1	0
## 52	50	1	0
## 53	50	0	1
## 54	49	1	0
## 55	50	0	1
## 56	49	0	1
## 57	89	1	0
## 58	69	0	1
## 59	69	1	0
## 60	89	1	0
## 61	36	0	1
## 62	89	1	0
## 63	80	0	1
## 64	93	0	1
## 65	92	1	0
## 66	89	1	0
## 67	79	1	0
## 68	59	0	1
## 69	94	0	1
## 70	33	0	1
## 71	100	1	0
## 72	90	0	1
## 73	89	1	0
## 74	99	1	0
## 75	88	1	0
## 76	88	0	1
## 77	79	0	1

## 78	48	0	1
## 79	79	1	0
## 80	100	0	1
## 81	78	0	1
## 82	27	1	0
## 83	32	0	1
## 84	47	1	0
## 85	30	1	0
## 86	45	0	1
## 87	48	0	1
## 88	39	1	0
## 89	43	0	1
## 90	69	1	0
## 91	48	0	1
## 92	59	1	0
## 93	53	1	0
## 94	54	0	1
## 95	59	0	1
## 96	50	1	0
## 97	29	1	0
## 98	32	0	1
## 99	30	1	0
## 100	39	1	0
## 101	50	0	1
## 102	50	0	1
## 103	58	0	1
## 104	40	1	0
## 105	53	1	0
## 106	38	0	1
## 107	28	1	0
## 108	34	1	0
## 109	65	0	1
## 110	39	0	1
## 111	39	0	1
## 112	49	1	0
## 113	63	0	1
## 114	61	0	1
## 115	53	1	0
## 116	48	1	0
## 117	51	0	1
## 118	58	1	0
## 119	55	1	0
## 120	51	0	1
## 121	49	0	1
## 122	78	0	1
## 123	33	1	0
## 124	19	1	0
## 125	78	0	1
## 126	9	1	0
## 127	64	1	0
## 128	50	0	1
## 129	67	1	0
## 130	44	1	0
## 131	54	0	1

## 132	59	1	0
## 133	56	0	1
## 134	43	0	1
## 135	70	1	0
## 136	63	0	1
## 137	63	1	0
## 138	43	1	0
## 139	39	0	1
## 140	54	1	0
## 141	67	0	1
## 142	67	0	1
## 143	77	1	0
## 144	66	0	1
## 145	66	1	0
## 146	59	1	0
## 147	54	0	1
## 148	57	0	1
## 149	55	1	0
## 150	65	0	1
## 151	54	1	0
## 152	64	0	1
## 153	78	1	0
## 154	59	1	0
## 155	58	0	1
## 156	60	0	1
## 157	63	1	0
## 158	62	0	1
## 159	71	1	0
## 160	68	0	1
## 161	81	0	1
## 162	83	0	1
## 163	92	0	1
## 164	71	1	0
## 165	69	1	0
## 166	59	1	0
## 167	91	1	0
## 168	100	0	1
## 169	89	0	1
## 170	78	0	1
## 171	76	0	1
## 172	75	1	0
## 173	72	1	0
## 174	86	1	0
## 175	75	1	0
## 176	82	0	1
## 177	37	1	0
## 178	53	0	1
## 179	63	0	1
## 180	63	1	0
## 181	65	0	1
## 182	64	1	0
## 183	63	0	1
## 184	54	1	0
## 185	69	1	0

## 186	74	0	1
## 187	58	0	1
## 188	63	1	0
## 189	63	0	1
## 190	50	1	0
## 191	67	0	1
## 192	85	1	0
## 193	58	0	1
## 194	58	0	1
## 195	69	1	0
## 196	69	1	0
## 197	59	0	1
## 198	88	1	0
## 199	68	1	0
## 200	59	0	1
## 201	50	0	1
## 202	58	0	1
## 203	39	1	0
## 204	53	1	0
## 205	58	0	1
## 206	54	1	0
## 207	55	1	0
## 208	59	0	1
## 209	48	0	1
## 210	59	1	0
## 211	60	1	0
## 212	57	1	0
## 213	52	0	1
## 214	53	0	1
## 215	60	1	0
## 216	64	0	1
## 217	68	0	1
## 218	90	1	0
## 219	88	1	0
## 220	75	1	0
## 221	78	0	1
## 222	70	0	1
## 223	93	1	0
## 224	79	0	1
## 225	73	0	1
## 226	89	1	0
## 227	100	1	0
## 228	89	0	1
## 229	100	0	1
## 230	100	1	0
## 231	100	0	1
## 232	100	1	0
## 233	79	0	1
## 234	81	1	0
## 235	82	1	0
## 236	73	0	1
## 237	78	0	1
## 238	81	1	0
## 239	78	0	1

## 240	81	1	0
## 241	69	0	1
## 242	64	1	0
## 243	66	1	0
## 244	64	0	1
## 245	78	0	1
## 246	71	1	0
## 247	66	0	1
## 248	72	1	0
## 249	96	0	1
## 250	86	1	0
## 251	100	1	0
## 252	100	0	1
## 253	79	0	1
## 254	54	1	0
## 255	93	0	1
## 256	85	1	0
## 257	47	0	1
## 258	67	0	1
## 259	69	1	0
## 260	69	1	0
## 261	69	0	1
## 262	69	1	0
## 263	79	1	0
## 264	78	0	1
## 265	68	0	1
## 266	39	0	1
## 267	60	1	0
## 268	73	1	0
## 269	78	0	1
## 270	30	1	0
## 271	69	1	0
## 272	66	0	1
## 273	19	0	1
## 274	9	1	0
## 275	5	1	0
## 276	14	1	0
## 277	27	1	0
## 278	30	0	1
## 279	48	0	1
## 280	69	0	1
## 281	42	0	1
## 282	37	1	0
## 283	51	1	0
## 284	63	1	0
## 285	67	1	0
## 286	87	0	1
## 287	86	0	1
## 288	87	0	1
## 289	69	0	1
## 290	58	1	0
## 291	58	1	0
## 292	59	0	1
## 293	58	0	1

## 294	58	0	1
## 295	54	1	0
## 296	54	1	0
## 297	66	0	1
## 298	79	1	0
## 299	79	1	0
## 300	81	0	1
## 301	86	0	1
## 302	80	0	1
## 303	88	1	0
## 304	91	1	0
## 305	99	1	0
## 306	50	0	1
## 307	100	0	1
## 308	100	0	1
## 309	100	0	1
## 310	100	1	0
## 311	100	1	0
## 312	100	1	0
## 313	68	1	0
## 314	78	0	1
## 315	70	0	1
## 316	72	0	1
## 317	78	0	1
## 318	77	1	0
## 319	81	1	0
## 320	78	1	0
## 321	68	0	1
## 322	47	0	1
## 323	39	1	0
## 324	68	0	1
## 325	49	1	0
## 326	39	1	0
## 327	20	0	1
## 328	59	1	0
## 329	59	0	1
## 330	69	0	1
## 331	50	1	0
## 332	67	0	1
## 333	59	1	0
## 334	50	1	0
## 335	69	0	1
## 336	50	1	0

Flow Location Scale Model

```
# Homoscedastic version, that only includes location model: Assumes that there is no difference in vari
flow_location_model <- brm(
  bf(flow_rating ~ shared_condition_order + chat_number + shared_prompt_order +
    (0 + condition_hai + condition_hh | participant_ID) +
    (1 | dyad_ID)),
```

```

data = df,
family = gaussian(),
cores = 4,
seed = 42,
iter = 4000,
control = list(adapt_delta = 0.95),
file_refit = "always",
file = "../Models/flow_location",
save_pars = save_pars(all = TRUE)
)

```

```
## Compiling Stan program...
```

```
## Trying to compile a simple C file
```

```
## Running /Library/Frameworks/R.framework/Resources/bin/R CMD SHLIB foo.c
```

```
## using C compiler: 'Apple clang version 17.0.0 (clang-1700.6.4.2)'
```

```
## using SDK: 'MacOSX26.0.sdk'
```

```
## clang -arch arm64 -std=gnu2x -I"/Library/Frameworks/R.framework/Resources/include" -DNDEBUG -I"/Us
```

```
## In file included from <built-in>:1:
```

```
## In file included from /Users/arichter/Library/R/arm64/4.5/library/StanHeaders/include/stan/math/prim
```

```
## In file included from /Users/arichter/Library/R/arm64/4.5/library/RcppEigen/include/Eigen/Dense:1:
```

```
## In file included from /Users/arichter/Library/R/arm64/4.5/library/RcppEigen/include/Eigen/Core:19:
```

```
## /Users/arichter/Library/R/arm64/4.5/library/RcppEigen/include/Eigen/src/Core/util/Macros.h:679:10: f
```

```
## 679 | #include <cmath>
```

```
##      |          ^~~~~~
```

```
## 1 error generated.
```

```
## make: *** [foo.o] Error 1
```

```
## Start sampling
```

```
## Warning: There were 90 divergent transitions after warmup. See
```

```
## https://mc-stan.org/misc/warnings.html#divergent-transitions-after-warmup
```

```
## to find out why this is a problem and how to eliminate them.
```

```
## Warning: Examine the pairs() plot to diagnose sampling problems
```

```
# Heteroscedastic version, that includes location and scale model: Directly models variability
```

```
flow_location_scale_model <- brm(
```

```
  bf(flow_rating ~ shared_condition_order + chat_number + shared_prompt_order +
```

```
    (0 + condition_hai + condition_hh | participant_ID) +
```

```
    (1 | dyad_ID),
```

```
  sigma ~ shared_condition_order + (1 | participant_ID)),
```

```
data = df,
```

```
family = gaussian(),
```

```
cores = 4,
```

```
seed = 42,
```

```
iter = 4000,
```

```
control = list(adapt_delta = 0.95),
```

```
file_refit = "always",
```

```
file = "../Models/flow_location_scale",
```

```
save_pars = save_pars(all = TRUE)
```

```
)
```

```

## Compiling Stan program...
## Trying to compile a simple C file

## Running /Library/Frameworks/R.framework/Resources/bin/R CMD SHLIB foo.c
## using C compiler: 'Apple clang version 17.0.0 (clang-1700.6.4.2)'
## using SDK: 'MacOSX26.0.sdk'
## clang -arch arm64 -std=gnu2x -I"/Library/Frameworks/R.framework/Resources/include" -DNDEBUG -I"/Us
## In file included from <built-in>:1:
## In file included from /Users/arichter/Library/R/arm64/4.5/library/StanHeaders/include/stan/math/prim
## In file included from /Users/arichter/Library/R/arm64/4.5/library/RcppEigen/include/Eigen/Dense:1:
## In file included from /Users/arichter/Library/R/arm64/4.5/library/RcppEigen/include/Eigen/Core:19:
## /Users/arichter/Library/R/arm64/4.5/library/RcppEigen/include/Eigen/src/Core/util/Macros.h:679:10: f
## 679 | #include <cmath>
##      |          ~~~~~
## 1 error generated.
## make: *** [foo.o] Error 1
## n 2: Iteration: 2001 / 4000 [ 50%] (Sampling)
## Chain 4: Iteration: 2400 / 4000 [ 60%] (Sampling)
## Chain 2: Iteration: 2400 / 4000 [ 60%] (Sampling)
## Chain 1: Iteration: 2400 / 4000 [ 60%] (Sampling)
## Chain 3: Iteration: 2400 / 4000 [ 60%] (Sampling)
## Chain 2: Iteration: 2800 / 4000 [ 70%] (Sampling)
## Chain 1: Iteration: 2800 / 4000 [ 70%] (Sampling)
## Chain 2: Iteration: 3200 / 4000 [ 80%] (Sampling)
## Chain 4: Iteration: 2800 / 4000 [ 70%] (Sampling)
## Chain 3: Iteration: 2800 / 4000 [ 70%] (Sampling)
## Chain 2: Iteration: 3600 / 4000 [ 90%] (Sampling)
## Chain 2: Iteration: 4000 / 4000 [100%] (Sampling)
## Chain 2:
## Chain 2: Elapsed Time: 8.97 seconds (Warm-up)
## Chain 2: 3.925 seconds (Sampling)
## Chain 2: 12.895 seconds (Total)
## Chain 2:
## Chain 1: Iteration: 3200 / 4000 [ 80%] (Sampling)
## Chain 3: Iteration: 3200 / 4000 [ 80%] (Sampling)
## Chain 4: Iteration: 3200 / 4000 [ 80%] (Sampling)
## Chain 1: Iteration: 3600 / 4000 [ 90%] (Sampling)
## Chain 3: Iteration: 3600 / 4000 [ 90%] (Sampling)
## Chain 4: Iteration: 3600 / 4000 [ 90%] (Sampling)
## Chain 1: Iteration: 4000 / 4000 [100%] (Sampling)
## Chain 1:
## Chain 1: Elapsed Time: 8.275 seconds (Warm-up)
## Chain 1: 7.666 seconds (Sampling)
## Chain 1: 15.941 seconds (Total)
## Chain 1:
## Chain 3: Iteration: 4000 / 4000 [100%] (Sampling)
## Chain 3:
## Chain 3: Elapsed Time: 8.826 seconds (Warm-up)
## Chain 3: 7.6 seconds (Sampling)
## Chain 3: 16.426 seconds (Total)
## Chain 3:
## Chain 4: Iteration: 4000 / 4000 [100%] (Sampling)
## Chain 4:
## Chain 4: Elapsed Time: 7.884 seconds (Warm-up)

```

```

## Chain 4:          9.114 seconds (Sampling)
## Chain 4:          16.998 seconds (Total)
## Chain 4:

## Start sampling

# Model comparison: how does gof change when including scale model
flow_location <- add_criterion(flow_location_model, "loo", moment_match = TRUE, reloo = TRUE)

## No problematic observations found. Returning the original 'loo' object.

## Automatically saving the model object in '../Models/flow_location.rds'

flow_location_scale <- add_criterion(flow_location_scale_model, "loo", moment_match = TRUE, reloo = TRUE)

## Warning: Some Pareto k diagnostic values are too high. See help('pareto-k-diagnostic') for details.

## 2 problematic observation(s) found.
## The model will be refit 2 times.

##
## Fitting model 1 out of 2 (leaving out observation 70)

## Start sampling

##
## Fitting model 2 out of 2 (leaving out observation 281)

## Start sampling

## Automatically saving the model object in '../Models/flow_location_scale.rds'

loo_compare(flow_location, flow_location_scale)

##               elpd_diff se_diff
## flow_location_scale  0.0      0.0
## flow_location      -5.4      4.0

summary(flow_location_scale_model)

## Family: gaussian
## Links: mu = identity; sigma = log
## Formula: flow_rating ~ shared_condition_order + chat_number + shared_prompt_order + (0 + condition_h
##          sigma ~ shared_condition_order + (1 | participant_ID)
## Data: df (Number of observations: 336)
## Draws: 4 chains, each with iter = 4000; warmup = 2000; thin = 1;
##        total post-warmup draws = 8000
##
## Multilevel Hyperparameters:

```

```
## ~dyad_ID (Number of levels: 21)
##           Estimate Est.Error 1-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sd(Intercept)      7.64      4.14      0.48      15.83 1.00        678        1151
##
## ~participant_ID (Number of levels: 42)
##           Estimate Est.Error 1-95% CI u-95% CI Rhat
## sd(condition_hai)      23.20      3.17      17.64      30.09 1.00
## sd(condition_hh)       14.40      3.02       8.60      20.53 1.00
## sd(sigma_Intercept)     0.25      0.11       0.03       0.45 1.00
## cor(condition_hai,condition_hh) 0.43      0.19       0.02       0.75 1.00
##
##           Bulk_ESS Tail_ESS
## sd(condition_hai)      1934      3701
## sd(condition_hh)       1389      1991
## sd(sigma_Intercept)     922      1297
## cor(condition_hai,condition_hh) 1961      3136
##
## Regression Coefficients:
##           Estimate Est.Error 1-95% CI u-95% CI Rhat
## Intercept      40.44      4.20      31.96      48.53 1.00
## sigma_Intercept  2.76      0.07       2.62       2.88 1.00
## shared_condition_order1 -3.56      3.84     -11.36       4.16 1.00
## chat_number       2.44      0.42       1.63       3.26 1.00
## shared_prompt_order  1.25      0.78      -0.25       2.83 1.00
## sigma_shared_condition_order1 0.25      0.10       0.06       0.45 1.00
##
##           Bulk_ESS Tail_ESS
## Intercept      2575      4560
## sigma_Intercept 2712      4727
## shared_condition_order1 2004      3633
## chat_number     4835      5361
## shared_prompt_order 8688      6107
## sigma_shared_condition_order1 5517      5715
##
## Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
## and Tail_ESS are effective sample size measures, and Rhat is the potential
## scale reduction factor on split chains (at convergence, Rhat = 1).
```

Social Presence Location Scale Model

```
# Homoscedastic version, that only includes location model: Assumes that there is no difference in vari
sp_location_model <- brm(
  bf(social_presence_rating ~ shared_condition_order + chat_number + shared_prompt_order +
    (0 + condition_hai + condition_hh | participant_ID) +
    (1 | dyad_ID)),
  data = df,
  family = gaussian(),
  cores = 4,
  seed = 42,
  iter = 4000,
  control = list(adapt_delta = 0.95),
  file_refit = "always",
  file = "../Models/sp_location",
```

```

save_pars = save_pars(all = TRUE)
)

## Compiling Stan program...

## Trying to compile a simple C file

## Running /Library/Frameworks/R.framework/Resources/bin/R CMD SHLIB foo.c
## using C compiler: 'Apple clang version 17.0.0 (clang-1700.6.4.2)'
## using SDK: 'MacOSX26.0.sdk'
## clang -arch arm64 -std=gnu2x -I"/Library/Frameworks/R.framework/Resources/include" -DNDEBUG -I"/Us
## In file included from <built-in>:1:
## In file included from /Users/arichter/Library/R/arm64/4.5/library/StanHeaders/include/stan/math/prim
## In file included from /Users/arichter/Library/R/arm64/4.5/library/RcppEigen/include/Eigen/Dense:1:
## In file included from /Users/arichter/Library/R/arm64/4.5/library/RcppEigen/include/Eigen/Core:19:
## /Users/arichter/Library/R/arm64/4.5/library/RcppEigen/include/Eigen/src/Core/util/Macros.h:679:10: f
## 679 | #include <cmath>
##      |         ~~~~~~
## 1 error generated.
## make: *** [foo.o] Error 1

## Start sampling

## Warning: There were 2 divergent transitions after warmup. See
## https://mc-stan.org/misc/warnings.html#divergent-transitions-after-warmup
## to find out why this is a problem and how to eliminate them.

## Warning: Examine the pairs() plot to diagnose sampling problems

# Heteroscedastic version, that includes location and scale model: Directly models variability
sp_location_scale_model <- brm(
  bf(social_presence_rating ~ shared_condition_order + chat_number + shared_prompt_order +
    (0 + condition_hai + condition_hh | participant_ID) +
    (1 | dyad_ID),
    sigma ~ shared_condition_order + (1 | participant_ID)),
  data = df,
  family = gaussian(),
  cores = 4,
  seed = 42,
  iter = 4000,
  control = list(adapt_delta = 0.95),
  file_refit = "always",
  file = "../Models/sp_location_scale",
  save_pars = save_pars(all = TRUE)
)

## Compiling Stan program...
## Trying to compile a simple C file

## Running /Library/Frameworks/R.framework/Resources/bin/R CMD SHLIB foo.c
## using C compiler: 'Apple clang version 17.0.0 (clang-1700.6.4.2)'

```

```

## using SDK: 'MacOSX26.0.sdk'
## clang -arch arm64 -std=gnu2x -I"/Library/Frameworks/R.framework/Resources/include" -DNDEBUG -I"/Us
## In file included from <built-in>:1:
## In file included from /Users/arichter/Library/R/arm64/4.5/library/StanHeaders/include/stan/math/prim
## In file included from /Users/arichter/Library/R/arm64/4.5/library/RcppEigen/include/Eigen/Dense:1:
## In file included from /Users/arichter/Library/R/arm64/4.5/library/RcppEigen/include/Eigen/Core:19:
## /Users/arichter/Library/R/arm64/4.5/library/RcppEigen/include/Eigen/src/Core/util/Macros.h:679:10: f
## 679 | #include <cmath>
##      |         ^~~~~~
## 1 error generated.
## make: *** [foo.o] Error 1
## / 4000 [ 30%] (Warmup)
## Chain 1: Iteration: 1600 / 4000 [ 40%] (Warmup)
## Chain 3: Iteration: 1600 / 4000 [ 40%] (Warmup)
## Chain 2: Iteration: 1600 / 4000 [ 40%] (Warmup)
## Chain 4: Iteration: 1600 / 4000 [ 40%] (Warmup)
## Chain 1: Iteration: 2000 / 4000 [ 50%] (Warmup)
## Chain 1: Iteration: 2001 / 4000 [ 50%] (Sampling)
## Chain 3: Iteration: 2000 / 4000 [ 50%] (Warmup)
## Chain 3: Iteration: 2001 / 4000 [ 50%] (Sampling)
## Chain 2: Iteration: 2000 / 4000 [ 50%] (Warmup)
## Chain 2: Iteration: 2001 / 4000 [ 50%] (Sampling)
## Chain 4: Iteration: 2000 / 4000 [ 50%] (Warmup)
## Chain 4: Iteration: 2001 / 4000 [ 50%] (Sampling)
## Chain 2: Iteration: 2400 / 4000 [ 60%] (Sampling)
## Chain 3: Iteration: 2400 / 4000 [ 60%] (Sampling)
## Chain 1: Iteration: 2400 / 4000 [ 60%] (Sampling)
## Chain 2: Iteration: 2800 / 4000 [ 70%] (Sampling)
## Chain 4: Iteration: 2400 / 4000 [ 60%] (Sampling)
## Chain 2: Iteration: 3200 / 4000 [ 80%] (Sampling)
## Chain 3: Iteration: 2800 / 4000 [ 70%] (Sampling)
## Chain 1: Iteration: 2800 / 4000 [ 70%] (Sampling)
## Chain 2: Iteration: 3600 / 4000 [ 90%] (Sampling)
## Chain 4: Iteration: 2800 / 4000 [ 70%] (Sampling)
## Chain 3: Iteration: 3200 / 4000 [ 80%] (Sampling)
## Chain 2: Iteration: 4000 / 4000 [100%] (Sampling)
## Chain 2:
## Chain 2: Elapsed Time: 2.15 seconds (Warm-up)
## Chain 2: 1.39 seconds (Sampling)
## Chain 2: 3.54 seconds (Total)
## Chain 2:
## Chain 1: Iteration: 3200 / 4000 [ 80%] (Sampling)
## Chain 4: Iteration: 3200 / 4000 [ 80%] (Sampling)
## Chain 3: Iteration: 3600 / 4000 [ 90%] (Sampling)
## Chain 1: Iteration: 3600 / 4000 [ 90%] (Sampling)
## Chain 4: Iteration: 3600 / 4000 [ 90%] (Sampling)
## Chain 3: Iteration: 4000 / 4000 [100%] (Sampling)
## Chain 3:
## Chain 3: Elapsed Time: 2.114 seconds (Warm-up)
## Chain 3: 2.279 seconds (Sampling)
## Chain 3: 4.393 seconds (Total)
## Chain 3:
## Chain 1: Iteration: 4000 / 4000 [100%] (Sampling)
## Chain 1:

```

```

## Chain 1: Elapsed Time: 2.057 seconds (Warm-up)
## Chain 1: 2.666 seconds (Sampling)
## Chain 1: 4.723 seconds (Total)
## Chain 1:
## Chain 4: Iteration: 4000 / 4000 [100%] (Sampling)
## Chain 4:
## Chain 4: Elapsed Time: 2.292 seconds (Warm-up)
## Chain 4: 2.516 seconds (Sampling)
## Chain 4: 4.808 seconds (Total)
## Chain 4:

## Start sampling

sp_location <- add_criterion(sp_location_model, "loo", moment_match = TRUE, reloo = TRUE) # Refits the model

## No problematic observations found. Returning the original 'loo' object.

## Automatically saving the model object in '../Models/sp_location.rds'

sp_location_scale <- add_criterion(sp_location_scale_model, "loo", moment_match = TRUE, reloo = TRUE)

## Warning: Some Pareto k diagnostic values are too high. See help('pareto-k-diagnostic') for details.

## 9 problematic observation(s) found.
## The model will be refit 9 times.

##
## Fitting model 1 out of 9 (leaving out observation 10)

## Start sampling

##
## Fitting model 2 out of 9 (leaving out observation 33)

## Start sampling

##
## Fitting model 3 out of 9 (leaving out observation 52)

## Start sampling

##
## Fitting model 4 out of 9 (leaving out observation 80)

## Start sampling

##
## Fitting model 5 out of 9 (leaving out observation 113)

```

```
## Start sampling

##
## Fitting model 6 out of 9 (leaving out observation 165)

## Start sampling

##
## Fitting model 7 out of 9 (leaving out observation 177)

## Start sampling

##
## Fitting model 8 out of 9 (leaving out observation 267)

## Start sampling

##
## Fitting model 9 out of 9 (leaving out observation 273)

## Start sampling

## Automatically saving the model object in '../Models/sp_location_scale.rds'
```

```
loo_compare(sp_location, sp_location_scale)
```

```
##               elpd_diff se_diff
## sp_location_scale    0.0      0.0
## sp_location       -14.8      9.3
```

```
summary(sp_location_scale_model)
```

```
## Family: gaussian
## Links: mu = identity; sigma = log
## Formula: social_presence_rating ~ shared_condition_order + chat_number + shared_prompt_order + (0 + sigma ~ shared_condition_order + (1 | participant_ID)
## Data: df (Number of observations: 336)
## Draws: 4 chains, each with iter = 4000; warmup = 2000; thin = 1;
## total post-warmup draws = 8000
##
## Multilevel Hyperparameters:
## ~dyad_ID (Number of levels: 21)
##               Estimate Est.Error 1-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sd(Intercept)    5.79      3.59    0.30   13.47 1.01      827    2050
##
## ~participant_ID (Number of levels: 42)
##               Estimate Est.Error 1-95% CI u-95% CI Rhat
## sd(condition_hai)    25.77      3.26    20.04   32.71 1.00
## sd(condition_hh)     11.87      2.64     6.53   17.20 1.00
## sd(sigma_Intercept)   0.44      0.08     0.30    0.60 1.00
```

```
## cor(condition_hai,condition_hh)      0.38      0.21      -0.08      0.75 1.00
##                                     Bulk_ESS Tail_ESS
## sd(condition_hai)                    2050      3867
## sd(condition_hh)                     1935      2453
## sd(sigma_Intercept)                  2899      4832
## cor(condition_hai,condition_hh)      3182      3945
##
## Regression Coefficients:
##                                     Estimate Est.Error 1-95% CI u-95% CI Rhat
## Intercept                          58.35      3.78    50.92    65.72 1.00
## sigma_Intercept                     2.73      0.09     2.56     2.90 1.00
## shared_condition_order1             -3.63      4.19   -11.78     4.77 1.00
## chat_number                         1.22      0.35     0.54     1.93 1.00
## shared_prompt_order                 0.43      0.68    -0.85     1.79 1.00
## sigma_shared_condition_order1       0.40      0.11     0.19     0.62 1.00
##                                     Bulk_ESS Tail_ESS
## Intercept                          2468      4126
## sigma_Intercept                     3240      4687
## shared_condition_order1             1794      3163
## chat_number                         9625      6168
## shared_prompt_order                 11519     6583
## sigma_shared_condition_order1       4770      5460
##
## Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
## and Tail_ESS are effective sample size measures, and Rhat is the potential
## scale reduction factor on split chains (at convergence, Rhat = 1).
```

Affect Location Scale Model

```
# Homoscedastic version, that only includes location model: Assumes that there is no difference in vari.
affect_location_model <- brm(
  bf(panas_rating ~ shared_condition_order + chat_number + shared_prompt_order +
    (0 + condition_hai + condition_hh | participant_ID) +
    (1 | dyad_ID)),
  data = df,
  family = gaussian(),
  cores = 4,
  seed = 42,
  iter = 4000,
  control = list(adapt_delta = 0.95),
  file_refit = "always",
  file = "../Models/affect_location",
  save_pars = save_pars(all = TRUE)
)
```

```
## Compiling Stan program...
```

```
## Trying to compile a simple C file
```

```
## Running /Library/Frameworks/R.framework/Resources/bin/R CMD SHLIB foo.c
## using C compiler: 'Apple clang version 17.0.0 (clang-1700.6.4.2)'
```

```
## using SDK: 'MacOSX26.0.sdk'
## clang -arch arm64 -std=gnu2x -I"/Library/Frameworks/R.framework/Resources/include" -DNDEBUG -I"/Us
## In file included from <built-in>:1:
## In file included from /Users/arichter/Library/R/arm64/4.5/library/StanHeaders/include/stan/math/prim
## In file included from /Users/arichter/Library/R/arm64/4.5/library/RcppEigen/include/Eigen/Dense:1:
## In file included from /Users/arichter/Library/R/arm64/4.5/library/RcppEigen/include/Eigen/Core:19:
## /Users/arichter/Library/R/arm64/4.5/library/RcppEigen/include/Eigen/src/Core/util/Macros.h:679:10: f
## 679 | #include <cmath>
##      | ~~~~~~
## 1 error generated.
## make: *** [foo.o] Error 1
```

```
## Start sampling
```

```
# Heteroscedastic version, that includes location and scale model: Directly models variability
affect_location_scale_model <- brm(
  bf(panas_rating ~ shared_condition_order + chat_number + shared_prompt_order +
    (0 + condition_hai + condition_hh | participant_ID) +
    (1 | dyad_ID),
    sigma ~ shared_condition_order + (1 | participant_ID)),
  data = df,
  family = gaussian(),
  cores = 4,
  seed = 42,
  iter = 4000,
  control = list(adapt_delta = 0.95),
  file_refit = "always",
  file = "../Models/affect_location_scale",
  save_pars = save_pars(all = TRUE)
)
```

```
## Compiling Stan program...
```

```
## Trying to compile a simple C file
```

```
## Running /Library/Frameworks/R.framework/Resources/bin/R CMD SHLIB foo.c
## using C compiler: 'Apple clang version 17.0.0 (clang-1700.6.4.2)'
## using SDK: 'MacOSX26.0.sdk'
## clang -arch arm64 -std=gnu2x -I"/Library/Frameworks/R.framework/Resources/include" -DNDEBUG -I"/Us
## In file included from <built-in>:1:
## In file included from /Users/arichter/Library/R/arm64/4.5/library/StanHeaders/include/stan/math/prim
## In file included from /Users/arichter/Library/R/arm64/4.5/library/RcppEigen/include/Eigen/Dense:1:
## In file included from /Users/arichter/Library/R/arm64/4.5/library/RcppEigen/include/Eigen/Core:19:
## /Users/arichter/Library/R/arm64/4.5/library/RcppEigen/include/Eigen/src/Core/util/Macros.h:679:10: f
## 679 | #include <cmath>
##      | ~~~~~~
## 1 error generated.
## make: *** [foo.o] Error 1
## : 1600 / 4000 [ 40%] (Warmup)
## Chain 1: Iteration: 2000 / 4000 [ 50%] (Warmup)
## Chain 1: Iteration: 2001 / 4000 [ 50%] (Sampling)
## Chain 3: Iteration: 1600 / 4000 [ 40%] (Warmup)
## Chain 4: Iteration: 2000 / 4000 [ 50%] (Warmup)
## Chain 4: Iteration: 2001 / 4000 [ 50%] (Sampling)
```

```

## Chain 2: Iteration: 2000 / 4000 [ 50%] (Warmup)
## Chain 2: Iteration: 2001 / 4000 [ 50%] (Sampling)
## Chain 1: Iteration: 2400 / 4000 [ 60%] (Sampling)
## Chain 3: Iteration: 2000 / 4000 [ 50%] (Warmup)
## Chain 3: Iteration: 2001 / 4000 [ 50%] (Sampling)
## Chain 4: Iteration: 2400 / 4000 [ 60%] (Sampling)
## Chain 2: Iteration: 2400 / 4000 [ 60%] (Sampling)
## Chain 1: Iteration: 2800 / 4000 [ 70%] (Sampling)
## Chain 3: Iteration: 2400 / 4000 [ 60%] (Sampling)
## Chain 4: Iteration: 2800 / 4000 [ 70%] (Sampling)
## Chain 2: Iteration: 2800 / 4000 [ 70%] (Sampling)
## Chain 1: Iteration: 3200 / 4000 [ 80%] (Sampling)
## Chain 3: Iteration: 2800 / 4000 [ 70%] (Sampling)
## Chain 2: Iteration: 3200 / 4000 [ 80%] (Sampling)
## Chain 4: Iteration: 3200 / 4000 [ 80%] (Sampling)
## Chain 1: Iteration: 3600 / 4000 [ 90%] (Sampling)
## Chain 2: Iteration: 3600 / 4000 [ 90%] (Sampling)
## Chain 3: Iteration: 3200 / 4000 [ 80%] (Sampling)
## Chain 4: Iteration: 3600 / 4000 [ 90%] (Sampling)
## Chain 1: Iteration: 4000 / 4000 [100%] (Sampling)
## Chain 1:
## Chain 1: Elapsed Time: 2.801 seconds (Warm-up)
## Chain 1: 3.195 seconds (Sampling)
## Chain 1: 5.996 seconds (Total)
## Chain 1:
## Chain 2: Iteration: 4000 / 4000 [100%] (Sampling)
## Chain 2:
## Chain 2: Elapsed Time: 3.236 seconds (Warm-up)
## Chain 2: 2.933 seconds (Sampling)
## Chain 2: 6.169 seconds (Total)
## Chain 2:
## Chain 3: Iteration: 3600 / 4000 [ 90%] (Sampling)
## Chain 4: Iteration: 4000 / 4000 [100%] (Sampling)
## Chain 4:
## Chain 4: Elapsed Time: 3.111 seconds (Warm-up)
## Chain 4: 3.241 seconds (Sampling)
## Chain 4: 6.352 seconds (Total)
## Chain 4:
## Chain 3: Iteration: 4000 / 4000 [100%] (Sampling)
## Chain 3:
## Chain 3: Elapsed Time: 3.857 seconds (Warm-up)
## Chain 3: 3.202 seconds (Sampling)
## Chain 3: 7.059 seconds (Total)
## Chain 3:

## Start sampling

## Warning: There were 42 divergent transitions after warmup. See
## https://mc-stan.org/misc/warnings.html#divergent-transitions-after-warmup
## to find out why this is a problem and how to eliminate them.

## Warning: Examine the pairs() plot to diagnose sampling problems

```

```

options(future.globals.maxSize = 2000 * 1024^2) # increase to 2 GB, fix model too big
affect_location <- add_criterion(affect_location_model, "loo", moment_match = TRUE, reloo = TRUE)

## No problematic observations found. Returning the original 'loo' object.

## Automatically saving the model object in '../Models/affect_location.rds'

affect_location_scale <- add_criterion(affect_location_scale_model, "loo", moment_match = TRUE, reloo = TRUE)

## Warning: Some Pareto k diagnostic values are too high. See help('pareto-k-diagnostic') for details.

## 8 problematic observation(s) found.
## The model will be refit 8 times.

##
## Fitting model 1 out of 8 (leaving out observation 17)

## Start sampling

##
## Fitting model 2 out of 8 (leaving out observation 61)

## Start sampling

##
## Fitting model 3 out of 8 (leaving out observation 70)

## Start sampling

##
## Fitting model 4 out of 8 (leaving out observation 78)

## Start sampling

##
## Fitting model 5 out of 8 (leaving out observation 81)

## Start sampling

##
## Fitting model 6 out of 8 (leaving out observation 130)

## Start sampling

##
## Fitting model 7 out of 8 (leaving out observation 198)

## Start sampling

```

```
##
## Fitting model 8 out of 8 (leaving out observation 306)

## Start sampling

## Automatically saving the model object in '../Models/affect_location_scale.rds'

loo_compare(affect_location, affect_location_scale)

##               elpd_diff se_diff
## affect_location_scale    0.0     0.0
## affect_location       -31.4    10.8

summary(affect_location_scale_model)

## Warning: There were 42 divergent transitions after warmup. Increasing
## adapt_delta above 0.95 may help. See
## http://mc-stan.org/misc/warnings.html#divergent-transitions-after-warmup

## Family: gaussian
## Links: mu = identity; sigma = log
## Formula: panas_rating ~ shared_condition_order + chat_number + shared_prompt_order + (0 + condition_1
##          sigma ~ shared_condition_order + (1 | participant_ID)
## Data: df (Number of observations: 336)
## Draws: 4 chains, each with iter = 4000; warmup = 2000; thin = 1;
##        total post-warmup draws = 8000
##
## Multilevel Hyperparameters:
## ~dyad_ID (Number of levels: 21)
##          Estimate Est.Error 1-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sd(Intercept)      9.28      3.22    2.24    15.53 1.00    1365    1369
##
## ~participant_ID (Number of levels: 42)
##          Estimate Est.Error 1-95% CI u-95% CI Rhat
## sd(condition_hai)      15.05      2.43    10.75    20.27 1.00
## sd(condition_hh)       10.21      2.12     6.55    14.78 1.00
## sd(sigma_Intercept)     0.53      0.09     0.37     0.74 1.00
## cor(condition_hai,condition_hh) 0.87      0.09     0.65     0.99 1.00
##          Bulk_ESS Tail_ESS
## sd(condition_hai)      1992      4160
## sd(condition_hh)       1889      3004
## sd(sigma_Intercept)     2247      4016
## cor(condition_hai,condition_hh) 2039      3051
##
## Regression Coefficients:
##          Estimate Est.Error 1-95% CI u-95% CI Rhat
## Intercept      59.67      3.26    53.15    65.99 1.00
## sigma_Intercept  2.27      0.10     2.08     2.46 1.00
## shared_condition_order1 -0.44      1.69    -3.77     2.93 1.00
## chat_number       0.57      0.24     0.13     1.08 1.00
## shared_prompt_order  0.37      0.39    -0.43     1.11 1.00
## sigma_shared_condition_order1 0.14      0.11    -0.06     0.35 1.00
```

```
##                               Bulk_ESS Tail_ESS
## Intercept                    4479    5929
## sigma_Intercept              3118    4917
## shared_condition_order1      4623    5227
## chat_number                  5853    6125
## shared_prompt_order          11706    6156
## sigma_shared_condition_order1 7626    6438
##
## Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
## and Tail_ESS are effective sample size measures, and Rhat is the potential
## scale reduction factor on split chains (at convergence, Rhat = 1).

save.image("backup.RData")
```